

Project & Data Management Plan

“Training Models on Moral Dilemma: How Artificial Intelligence (AI) “thinks” about Morality”

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Abstract—Project plan and outline in order to Evaluating Moral Reasoning and cognition in AI

I. PROJECT OVERVIEW

A. Background and Related Research

This exploration takes from John Searle’s Chinese Room Argument (Searle, 1980), which stands to contest the assumption that genuine moral perception/cognition (semantics) versus imitation of pattern recognition through statistical association (syntax).

Early more modern parallels with goals of instilling ethical reasoning in AI was attempts with The Scruples data sets (Lourie, et al., 2021), with attempts to observe whether models were able to learn moral acceptability based on public/social consensus.

Similar grounded approach was further delved by (Hendrycks, et al., 2021) through the ETHICS dataset. Introducing significantly more structure with moral reasonings based on established philosophical frameworks like deontology, and utilitarianism.

Which there after leads to the, The Delphi model (Jiang, et al., 2021), on foundations from Hendrycks, was one of the pioneering large-scale attempts to build an AI system with offering not merely yes or no classifications that was done in the Hendrycks datasets, but also addressing open-ended responses.

These questions tackle the reliability for models to be able to replicate accurately moral concepts. However, lacking the ability to look under the hood that is the model and be known whether there is truly semantic understanding of morality or mere syntactic mimicry through statistical means.

B. Research Question & Objectives

This paper foremost explores **how machine learning models performance across absolute and relative morality**, where absolute ethics represent a fixed normative standard (i.e. deontology), relative ethics capture moral judgements that allow fluidity i.e. utilitarianism, emotivism. These comparisons give rise to whether models are able to generalise across differing frameworks or are only able to obey specific types of moral paradigms.

Furthermore, the study examines **the consistency of AI ethical judgments under different frameworks of similar dilemmas**. By rewriting and testing the stability of responses in changed contexts.

Extending each individual lines of questioning, an overarching aim to question ethico-cognition in AI. **how much these systems learn and understand on a principled level, as opposed to merely imitating surface forms of language** – a distinction that is motivated in part by John Searle’s (1980) Chinese Room Argument.

C. Main Contributions

This paper seeks to contribute both conceptually and practically. In the emerging field of artificial moral reasoning and AI transparency. Conceptually, it provides a framework for evaluating, through combining philosophical discourse with empirical model testing, AI systems demonstrate genuine moral understanding or merely mimic moral reasoning through statistical pattern recognition.

From a practical perspective, the project aims to reach measurable criteria and methods for assessing AI ethical reasoning. This includes assessment metrics for ethical consistency and justification quality, and exploring methods to examine how models internally encode ethical concepts. Through this examination, can inform the design of transparent and accountable AI systems that are able to align more closely with human ethical expectations.

Concrete results of this research include:

- A prototype evaluation framework to benchmark AI ethical reasoning across absolute and relative ethical frameworks.
- Empirical findings demonstrate how different training regimes (theory-based vs. imitation-based) affect moral generalization and reasoning consistency.
- A set of operational indicators to distinguish genuine understanding from statistical imitation – such as moral coherence, causal reasoning and counterfactual robustness.
- Recommendations for incorporating ethical reasoning modules into AI governance, education, and policy evaluation systems.

II. PROJECT PLAN

A. Task list:

Stages	TASKS	DAYS ALLOCATED
Project Scoping	Finalize research question and scope	10
& Background	Conduct literature review on AI + morality, datasets	15
Research	Literature review on model selection	15

	Define evaluation metrics and success criteria	8
Dataset Design and Cleanup	Choose dataset focus (cross-cultural, uncertainty, etc.)	14
	Combining Dataset	7
	Cleaning up Data set	5
	Preliminary data set exploration	5
Model Selection & setup	Choose baseline models (rule-based, shallow)	5
	Choose transformer models (LLaMA, GPT, Mistral, etc.)	7
	Set up training environment (Python, GPUs, libraries)	37
	Preprocess dataset and tokenization	10
	Fine-tune baseline models on ETHICS	12
	Fine-tune models on your custom dataset	14
	Sanity-check and validate model outputs	7
Evaluation	Define and calculate evaluation metrics (accuracy, robustness, bias)	2
	Conduct interpretability analysis (attention, embeddings, interventions)	14
	Cross-cultural or moral uncertainty tests	15
Analysis & Reflection	Relate findings to philosophical framing (Chinese Room, etc.)	9
	Identify limitations, biases, implications	7
Write-up & Presentation Prep	Draft report/paper (intro, method, results)	18
	Create figures, charts (attention maps, embeddings)	15
	Revise, finalize, and prepare presentation	14

Fig. 1 List of task and key milestones with estimated allocated time

B. Timeline:

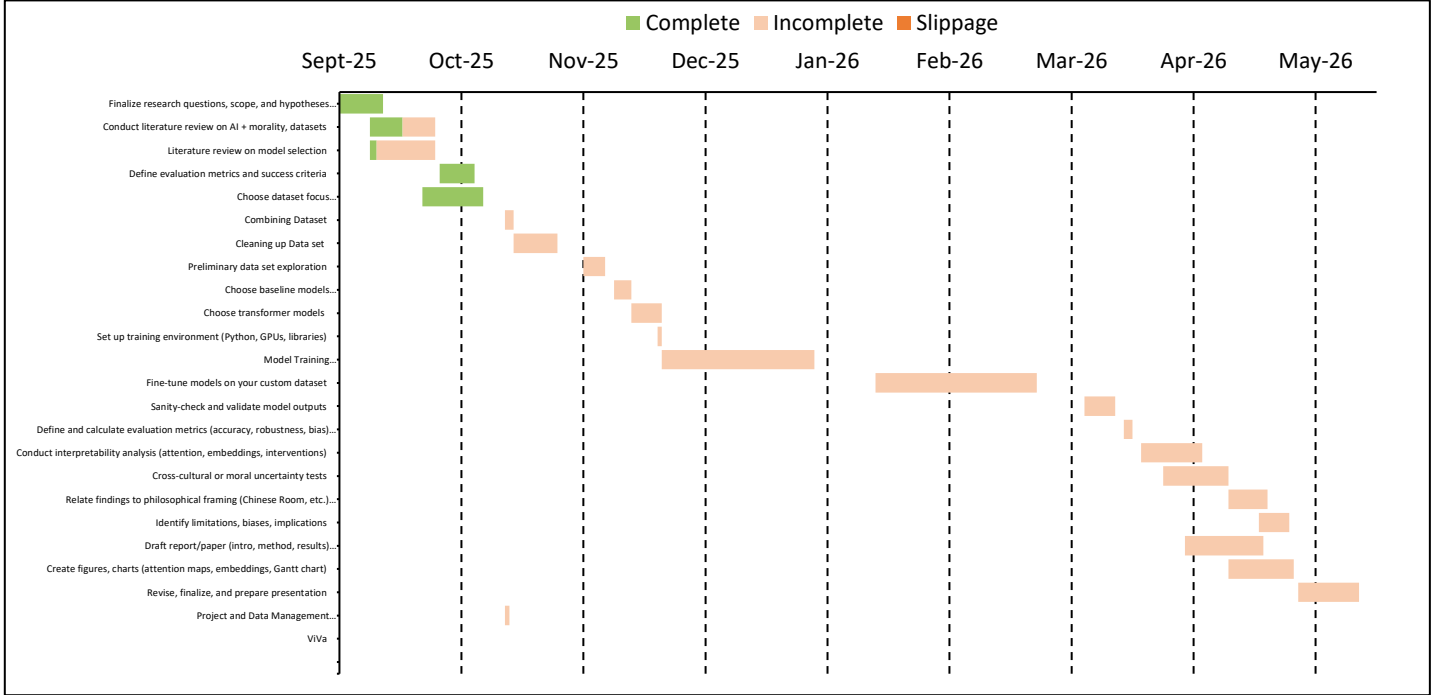


Fig. 2 Overview of Gantt outlining project timeline on 30th Oct 2025

III. DATA MANAGEMENT PLAN

A. Chosen Datasets

DATASET	LINK
ETHICS	https://github.com/hendrycks/ethics (Hendrycks, et al., 2021)
MoRe Bench	https://github.com/morebench/morebench (Chiu, et al., 2025)

ETHICS was chosen for its pioneering grounding design encapsulating many different ethical frameworks. However, its annotation and classification are binary which is where, though on the flip side of the coin, the **MoReBench** was likewise selected for its pluralistic moral reasoning, and incorporating human-annotated explanation. the MoReBench dataset allows for more in-depth evaluation.

The MoReBench dataset shares similar moral frameworks as well allowing both datasets to be concatenated together for more dimension in analysis.

1) ETHICS Dataset

a) Overview

The ETHICS dataset was one of the first large-scale, structured attempts to introduce philosophical rigor into moral reasoning datasets.

b) Metadata & Structure:

ATTRIBUTE	DESCRIPTION
Data Format	.json, .csv, .txt
Size	~33 MB
Records	~130,000 text scenarios
Annotation	Binary classification (acceptable = 1, unacceptable = 0)
Moral Subdomains	Deontology, Utilitarianism, Virtue Ethics, Commonsense Morality, Justice
Source / Origin	Created by UC Berkeley, UCSD, and Google Research (ICLR 2021)
Repository	https://github.com/hendrycks/ethics
Data Splits	Pre-divided into training, validation, and test sets

Fig. 3 Table metadata descript of ETHICS dataset

2) MoRe Bench Dataset

a) Overview

MoReBench represent more recent attempts (having been published in 2025) for a comprehensive large-scale effort to access procedural and pluralistic moral reasoning in models.

The interdisciplinary team in UC Berkeley, Stanford University seeks to test how well AI systems can justify morality across diverse moral frameworks beyond binary outcomes.

b) Metadata & Structure:

ATTRIBUTE	DESCRIPTION
Data Format	.json, .csv, .txt
Size	~200 MB
Records	~ 50 000 human-authored moral scenarios and justification pairs
Annotation	Multi-labelled open-ended annotations with reasoning
Moral Subdomains	Deontology, Utilitarianism, Virtue Ethics, Justice, Care ethics
Source / Origin	Created by UC Berkeley, UCSD, and Google Research (ICLR 2021)
Repository	https://github.com/morebench/morebench
Data Splits	Pre-divided into training, validation, and test sets

Fig. 4 Table metadata descript of ETHICS dataset

c) Datasets Ethical Compliance:

CRITERION	ETHICS	MORE BENCH
GDPR Compliance	Dataset is text-only, without any personally identifiable information	Anonymized textual data with no personal information
UH Ethical Policies	Data anonymized under MIT license	Secondary data anonymized, collected under informed consent
Permission to Use	Used and open for research under MIT license	-
Original Data Ethics	Created By researchers from UC Berkeley, google	-

Fig. 5 Ethical check for both ETHICS and MoReBench dataset

B. Version Control

- GitHub Repository: <https://github.com/nujgnil/AI-ethics>
- Commit Frequency: Daily updates to capture progress
- Versioning System:
 - Scripts: Tracked and named using the
 - Data Storage and Versioning
 - Data predicted to be possibly larger than what standard GitHub permits.
 - Commits: Each commit with relation to stage of research (i.e. feat: preprocessing script, etc...)

C. Security & Storage

- Primary Storage: Personal Machine
- Secondary Backup: GitHub (auto-sync).
- Backup Frequency: Daily.

IV. REFERENCES

- Chiu, Y. et al., 2025. *MoReBench: Evaluating Procedural and Pluralistic Moral Reasoning in Language Models, More than Outcomes*, Online: arXiv / Cornell University Library.
- Hendrycks, D. et al., 2021. *Aligning AI with Shared Human Values*. online , ICLR.
- Jiang, L. et al., 2021. Can machines learn morality? The Delphi experiment. *arXiv preprint arXiv:2110.07574*, -(-), pp. -.
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- Lourie, N., Le Bras, R. & Choi, Y., 2021. *Scruples: A Corpus of Community Ethical Judgments on 32,000 Real-Life Anecdotes*. Online, AAAI Press.
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